

5.1 Eigenvalues and Characteristic Polynomials

An eigenvalue of a linear transformation represented by a square matrix A is a scalar λ such that there exists a non-zero vector x satisfying the fundamental equation $Ax = \lambda x$. The vector x is termed an eigenvector corresponding to λ .

To determine the eigenvalues, we rewrite the matrix equation as $(A - \lambda I)x = 0$. Non-trivial solutions exist if and only if the determinant $\det(A - \lambda I)$ vanishes. This scalar equation defines the characteristic polynomial of degree n .

In geometric terms, an eigenvector points in an invariant direction that undergoes simple dilation under the linear transformation, with the dilation factor given precisely by the associated eigenvalue.

5.2 Orthogonal Diagonalization & Symmetric Matrices

A matrix S is symmetric if S equals its transpose S^T . The Spectral Theorem guarantees that every real symmetric matrix has exclusively real eigenvalues and possesses an orthonormal basis of eigenvectors.

Consequently, any real symmetric matrix S can be decomposed as $S = Q \Lambda Q^T$, where Q is an orthogonal matrix whose columns are the normalized eigenvectors, and Λ is a diagonal matrix containing the eigenvalues.

This spectral decomposition is of paramount importance in principal component analysis (PCA), where data covariance matrices are diagonalized to extract principal axes of variance.

5.3 Singular Value Decomposition (SVD)

Singular Value Decomposition generalizes spectral diagonalization to arbitrary rectangular $m \times n$ matrices. For any real matrix A , there exist orthogonal matrices U and V such that $A = U \Sigma V^T$.

The singular values σ_i along the diagonal of Σ are the square roots of the eigenvalues of $A^T A$. They measure the geometric stretch along the principal semi-axes of the transformed unit sphere.

SVD directly powers pseudoinverse computation, low-rank matrix approximation via the Eckart-Young-Mirsky theorem, and latent semantic indexing in computational linguistics.